

UNISPHERE CORP

AI SALES SCRIPT AUTOMATION

An internal AI production platform that turns a marketing hypothesis into a campaign-ready script and launch-ready creative set — built on a 5-stage AI writing pipeline, human-in-the-loop review with full version history, 4-role RBAC, and a 3-mode Image Studio, and shipped end to end from proof-of-concept discovery to a production deployment on Google Cloud Run.

5-Stage AI Writing Pipeline	4-Role RBAC & Team Management
3-Mode Image Studio	Google Cloud Run Deployment

CLIENT

Unisphere Corp (formerly PROTUS)

CATEGORY

AI Sales Ops · Internal Tooling · GenAI

STACK

FastAPI · React/Vite · Postgres (Neon) · OpenRouter

SCOPE

Full build · 5-stage AI pipeline · RBAC · Image Studio · Deployment

OVERVIEW

From a read-only PoC walkthrough to a production script factory.

Unisphere Corp (originally scoped and built under the working name PROTUS) is an internal AI production platform that takes a marketing hypothesis through a 5-stage AI writing pipeline, a human review and rewrite workflow with full version history, an approval gate, and on into a 3-mode Image Studio for campaign creative — with an optional Notion handoff and cost tracked separately for script production and image generation throughout.

5 AI agent stages per script — Writer, Copy Chief, Auditor, Reconciler, Revision Writer	4 RBAC roles with a dedicated Team management page
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16

Client-curated, test-verified models in the writer catalogue

3

Image Studio generation modes, each with versioning

What The Platform Had To Do

- ✓ **Freeze inputs:** every run's brief, competitor script, control breakdown, and avatar research locked into an immutable snapshot at creation.
- ✓ **Pipeline the writing:** fan each run out across script variants and push every one through the same 5-stage chain.
- ✓ **Keep a human in the loop:** reviewers rewrite, comment, and compare versions before anything is approved.
- ✓ **Close the loop to creative:** an approved script becomes a Launch Package and can generate campaign images without leaving the app.

PROJECT ORIGIN

Reverse-engineered from a vague brief, not built from a finished spec.

Before any code was written, the team ran a read-only walkthrough of the client's own proof-of-concept tool ("Writer Flow PoC," Centaurtea) to reverse-engineer the intended product. That exploration established the mental model the whole build is based on, and produced the open-questions list — including the absence of confirmed auth and roles in the client's own tool — that shaped every subsequent conversation and became one of this build's own headline features.

Mental Model Established From The PoC Walkthrough

- ✓ **A run is a marketing experiment:** shared inputs frozen into an immutable snapshot, fanned out across variants (Clean/Control, Iteration A/B/C).
- ✓ **Each variant is pipelined:** every variant is pushed through the multi-stage AI writing pipeline independently.
- ✓ **A human decides:** pipeline output is handed to a reviewer for review, scoring, and approval before anything ships.
- ✓ **Approval unlocks Launch:** approval feeds a Launch Package stage — naming, asset/image generation, and handoff.

The result was a clean, linear flow that the application follows today:

Create → Monitor → Review → Launch → Admin

Build Phases At A Glance

Phase	What Shipped
Milestone 1	Core pipeline: runs, variants, 5-agent writer chain, hybrid Postgres + filesystem storage, JWT auth, basic roles.
Model curation	Replaced the open-ended model picker with a client-curated, tier-colour-coded, test-verified 16-model catalogue.
Launch Packages + Image Studio (Mode 1)	Approved-script list → competitor-image analysis → prompt enhancement → OpenRouter image generation → Assets Outbox auto-feed.
Production hardening	GCS-backed image storage, Notion integration, RBAC expanded to 4 roles, cost split by category.
Metadata & naming-convention system	Brief-only extraction of filename/URL metadata at run setup, editable Script IDs, deterministic filename assembly.
Script-centric reorg	App reorganized around the script as the unit of work: Compare Scripts diff tool, script-level Product URL, live Notion schema migration, editable Run ID.
Reliability pass	Root-caused and fixed a silent script-truncation bug, a CORS-on-error infra bug, and several race-condition / re-assembly data-loss bugs.
Image Studio Modes 2 & 3	Manual Scene Builder and script-driven batch image generation with categories, versioning, and image-editing.
QA hardening batch	Run ID validation, preset bugs, atomicity fixes, layout overflow fixes, version-history visibility fixes.

ARCHITECTURE

A run becomes a script, a script becomes a launch package.

The platform is organised around a single lifecycle: a run is created through a guided wizard, its inputs are frozen, its variants are pushed through the AI writing pipeline, a human reviews and approves the output, and the approved script flows into a Launch Package and, optionally, Image Studio and Notion. Later in the build this model shifted from run-centric to script-centric, so that the script — not the run — became the consistent unit of work across every surface of the app.

LAYER	RESPONSIBILITY	TECHNOLOGY
Run Creation	Guided wizard captures Run Setup, Inputs, Variations, and Review & Submit; freezes six input fields into an immutable snapshot.	React / Vite wizard
AI Pipeline	Fans each variant through a 5-stage agent chain, one OpenRouter LLM call per stage, with retry/regenerate.	FastAPI, OpenRouter
Review & Versioning	Text-selection rewrite comments, independent version history, and a hand-rolled Compare Scripts diff engine.	React, custom diff engine
Metadata & Naming	Single structured-output call extracts filename/URL metadata; lookup tables resolve deterministic filenames.	FastAPI, Postgres
Launch & Assets	Approved-script list, Assets Outbox auto-population, and image-generation lock rules.	React, FastAPI
Image Studio	Three generation modes sharing one master prompt template, with versioned, editable images.	OpenRouter image models
Integrations	Per-script Notion pages via the 3-step File Upload API; GCS-backed image storage.	Notion API, Google Cloud Storage
Platform	JWT auth, 4-role permission matrix, dark/light mode, toast notifications.	JWT, bcrypt, React

Core Infrastructure

- ✓ **Auth:** JWT-based login using bcrypt directly (no passlib), for bcrypt 5.x compatibility.
- ✓ **Permissions:** single source-of-truth matrix (7 permissions × 4 roles), enforced server-side and mirrored on the frontend so the UI gates purely off server-returned permissions.
- ✓ **Storage:** hybrid Postgres + filesystem at first, later GCS-backed image storage once Cloud Run's ephemeral filesystem made that necessary.
- ✓ **Deployment:** Dockerized FastAPI on Google Cloud Run, GitHub Actions CI/CD using Workload Identity Federation — no long-lived service-account keys.

The 5-Stage AI Writing Pipeline

Each selected variant runs the same chain, one OpenRouter LLM call per stage, ending in the version that enters human review.

1 Writer	Produces the first draft under one of three modes — Adapt (competitor-aware adaptation), Iterate (minimal-diff edit of an approved Control), or Rewrite (rebuilt around the core problem/solution).
↓	
2 Copy Chief	Strengthens persuasion, structure, and flow while preserving the core message.
↓	
3 Auditor	Runs an independent quality pass, flagging weaknesses ahead of reconciliation.
↓	
4 Reconciler	Resolves conflicts between the Writer's output and the Auditor's findings into one clean set of recommendations.
↓	
5 Revision Writer	Applies the reconciled recommendations and produces the final version that enters the human review queue.

Every system prompt in this chain is editable Markdown, versioned through the global prompts mechanism used across the app — and every candidate model in the 16-model catalogue was fired against OpenRouter with a real completion before being wired in, with 4 of the original 20 candidates excluded as invalid or perpetually rate-limited.

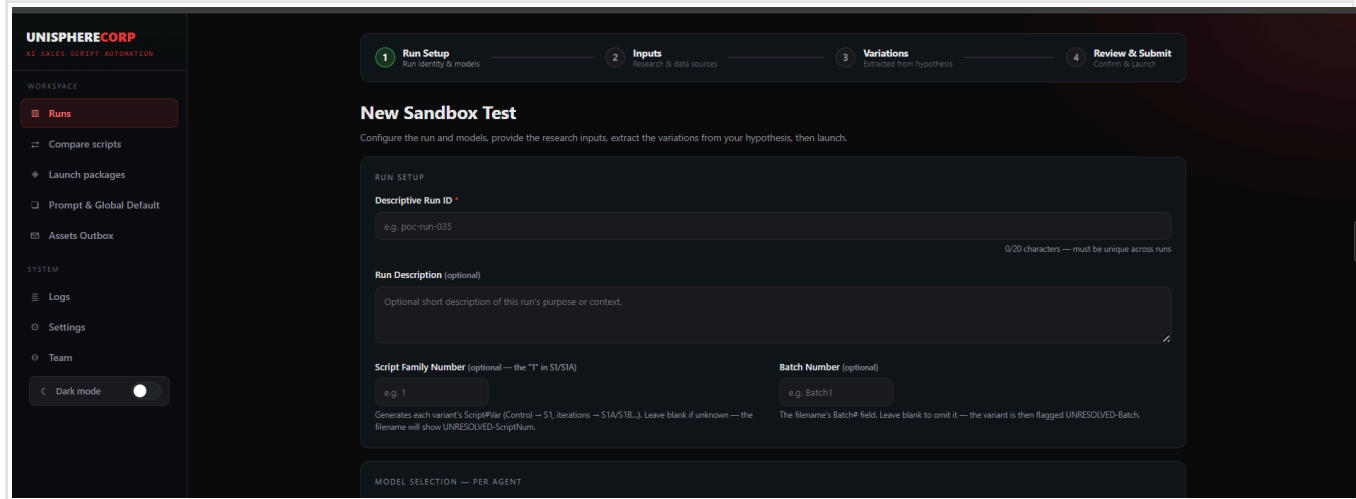
FROM THE PRODUCT

Ten screens that show the platform actually running.

The sessions below evidence the completed work across the full project — real screens from the deployed application, not mockups.

Run Creation & The Model Catalogue

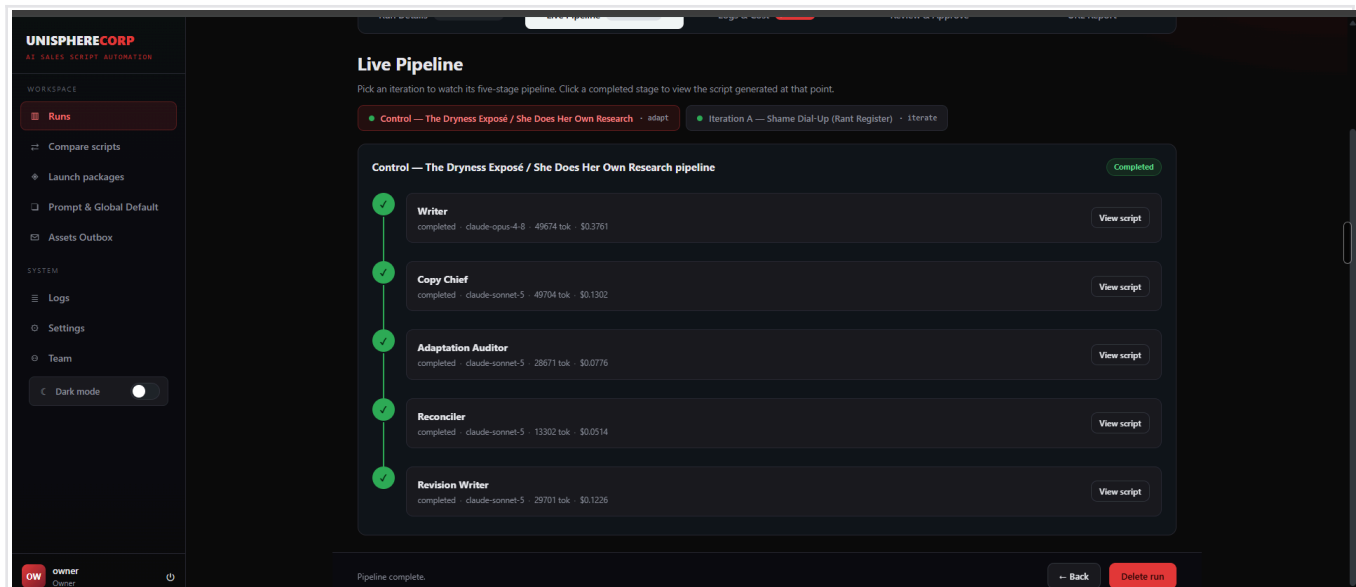
The guided wizard — Run Setup → Inputs → Variations → Review & Submit — and the client-curated, tier-coloured 16-model catalogue behind it.



New Sandbox Test — Run Setup step of the guided wizard, with the 4-step progress rail and per-agent model selection.

The Live Pipeline

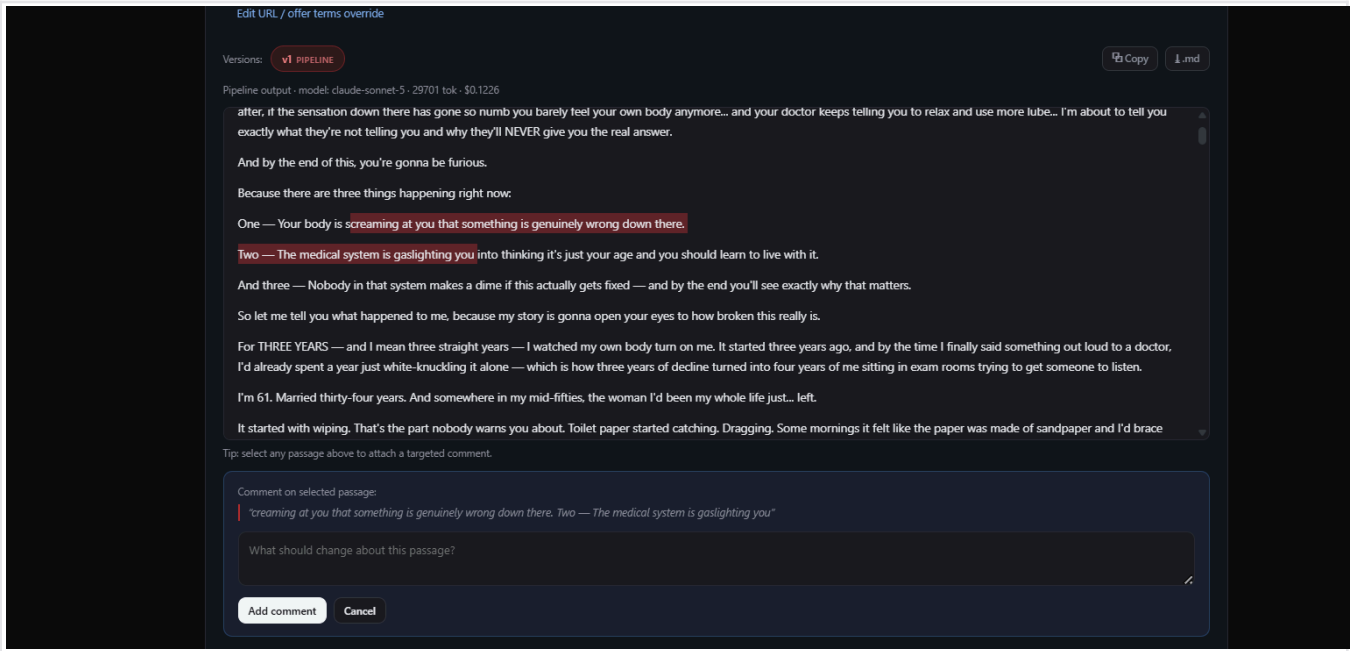
A completed 5-stage run — Writer through Revision Writer — with per-stage model, token count, and cost visible at a glance.



Live Pipeline view for a Control variant: all five stages completed, each showing model, token usage, and cost.

Human Review, Rewrite & Comments

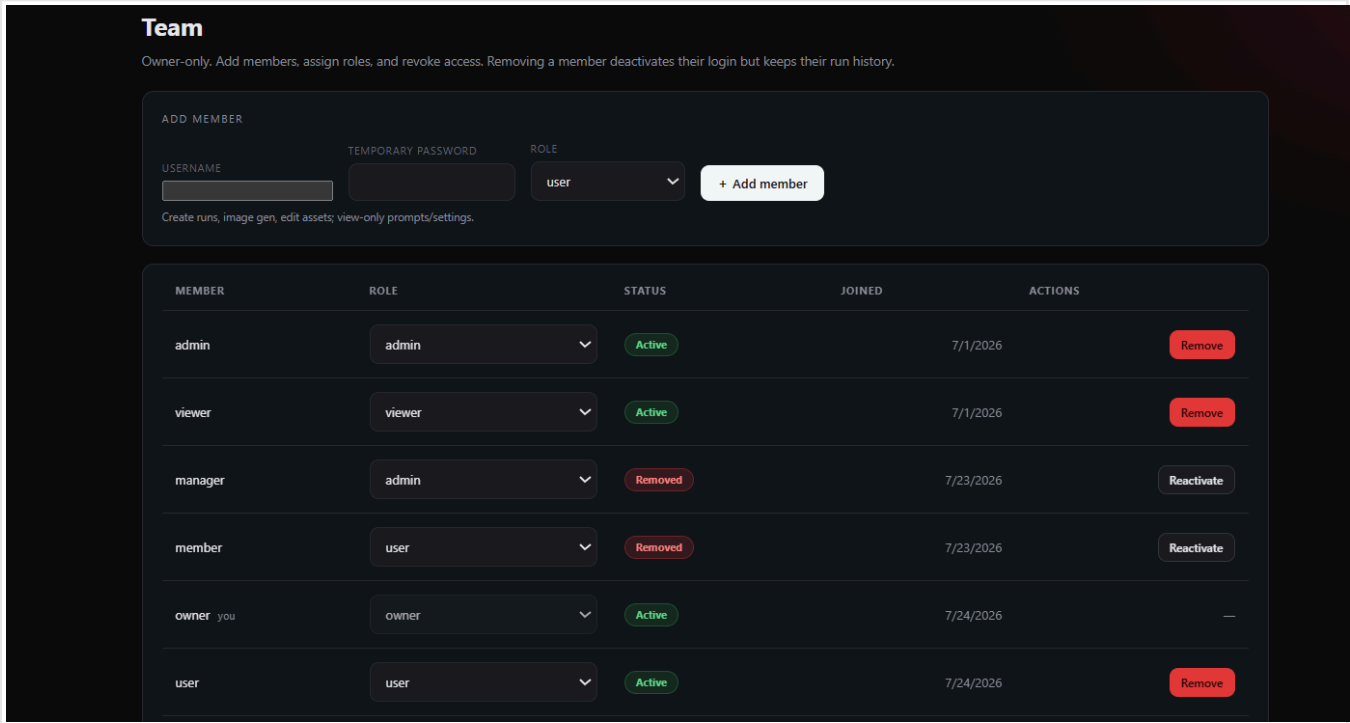
Reviewers select any passage, attach a targeted comment, and trigger a rewrite — every rewrite inserts a new version rather than overwriting the last.



Reviewer workspace: a highlighted passage with an attached rewrite comment, ready to submit against the pipeline output.

RBAC & Team Management

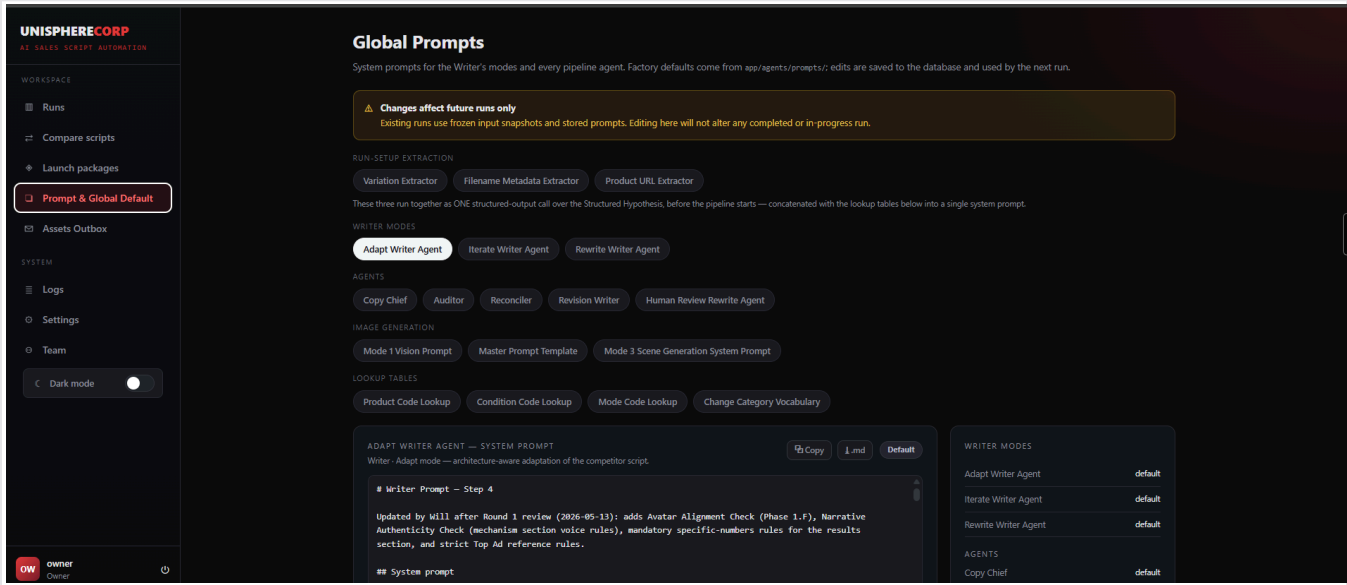
The Owner-only Team page: add members, assign one of four roles, deactivate accounts, and reactivate them later without losing run history.



Team page showing active and removed members across the Owner / Admin / User / Viewer role set.

Global Prompts & Settings

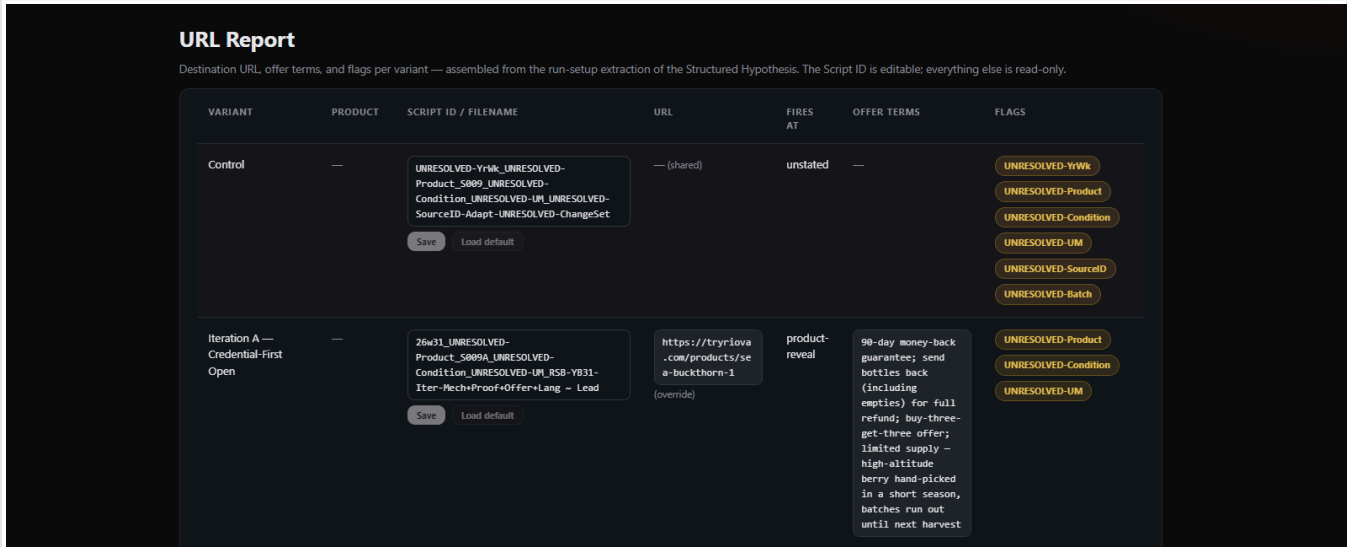
Every agent system prompt — 5 writer-chain agents, 3 writer-mode variants, extraction prompts, and 3 Image Studio prompts — is editable Markdown with Customized/Default badges.



Global Prompts screen: prompt categories in the sidebar, with the Adapt Writer Agent system prompt open for editing.

Metadata & Naming-Convention System

A single structured-output call extracts product, condition, mechanism, and per-variation URL/offer/flags at run-setup time; the Script ID is the one operator-editable field in an otherwise read-only panel.



URL Report: destination URL, offer terms, and resolution flags per variant, assembled from the run-setup extraction.

Image Studio — Script-Driven Batch

Mode 3 lets a reviewer pick a batch of images, assign each a category, and let an LLM draft one editable scene per image from the approved script before generating.

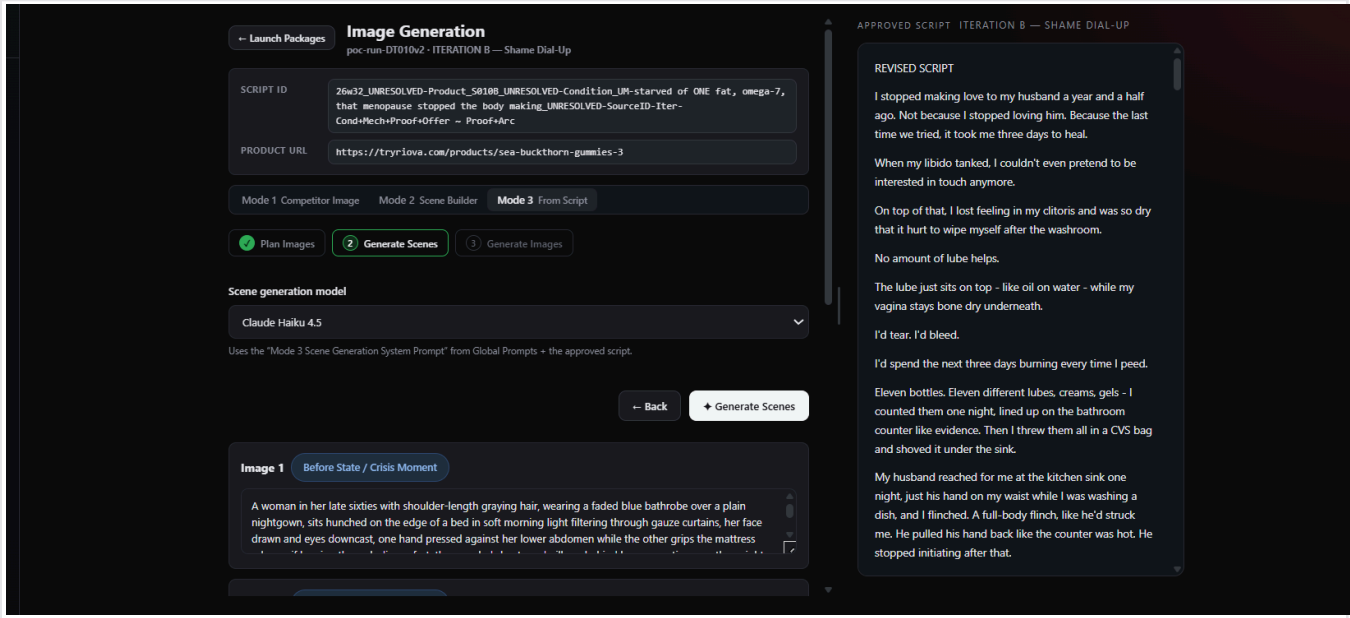
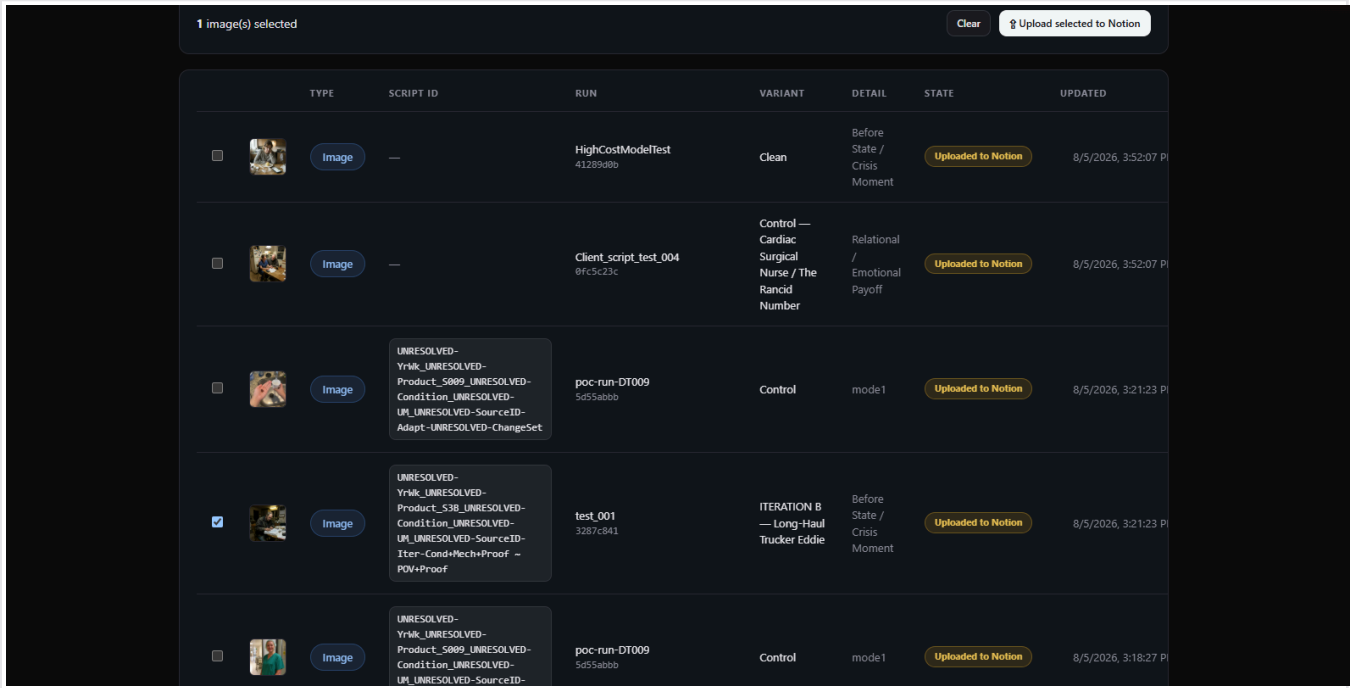


Image Studio, Mode 3: scene-generation step with the approved script visible alongside the drafted scene description.

Assets Outbox

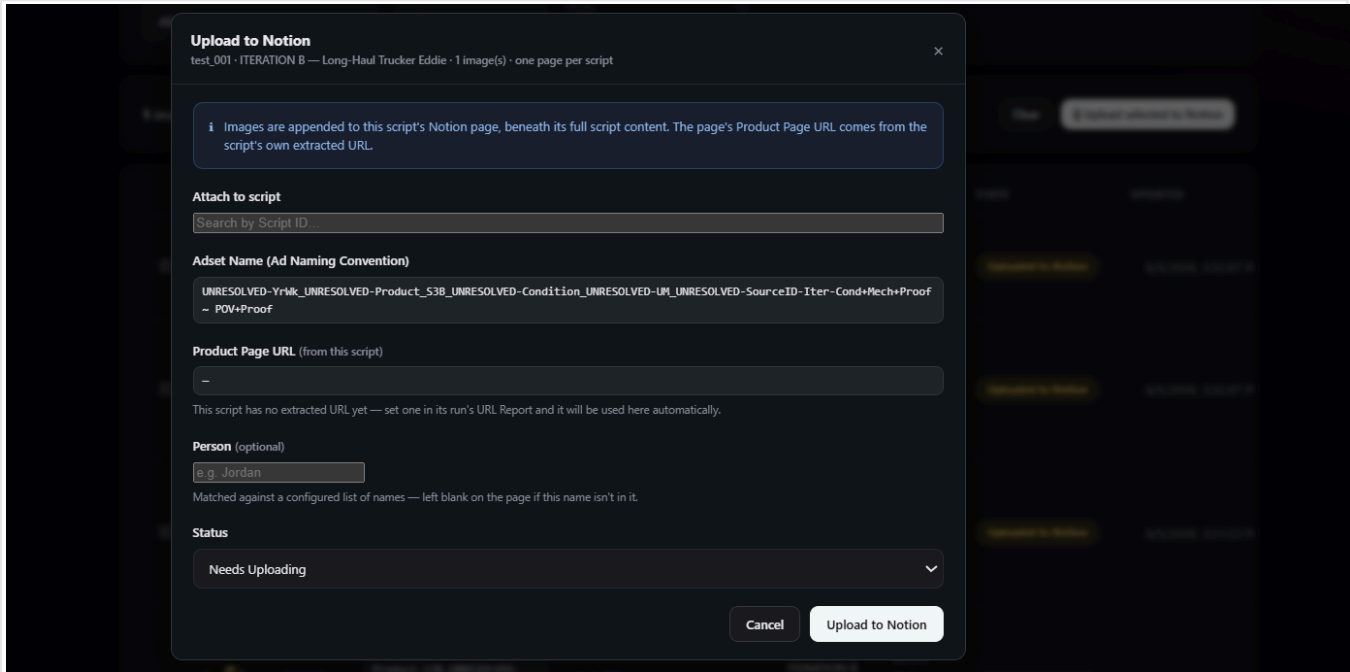
Every completed script and generated image lands here automatically — filterable by run, type, and date range, with multi-select batch push to Notion.



Assets Outbox: generated images with their source run, variant, and Notion upload status.

Notion Integration

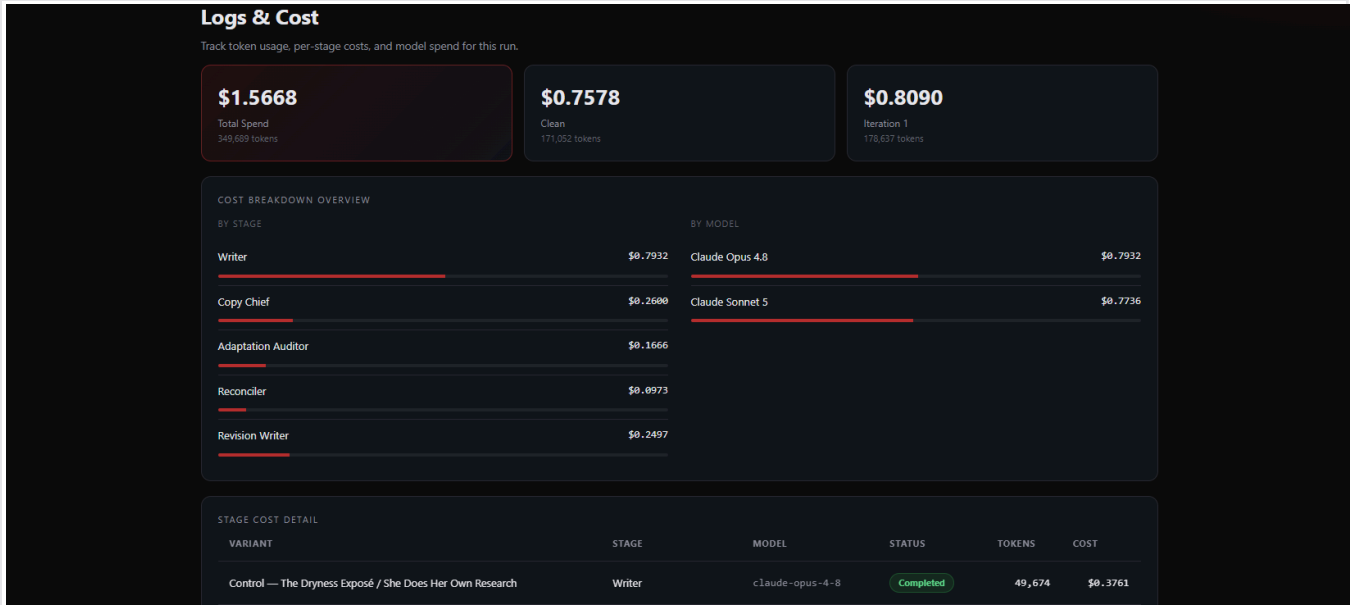
A manual, per-asset “Send to Notion” action appends images to that script's own Notion page as real embedded file blocks via Notion's 3-step File Upload API.



Upload to Notion dialog: adset naming convention, product URL, and status prior to sending images to the script's page.

Dashboard — Logs & Cost

Cost is tracked per stage, per model, and per variant, and reported as two genuinely separate categories: script production spend and image generation spend.



Logs & Cost view for a run: total spend, per-stage cost breakdown, and per-model cost breakdown.

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AI SALES SCRIPT AUTOMATION

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Runs

Home base for AI script generation tests. Start a new sandbox, resume an in-progress run, or open completed runs for review.

61

Total Runs

133

Need Reviews

20

Approved Scripts

\$56.3511

Script Production Cost

\$2.8271

Image Generation Cost

Search projects...

AllRunningCompletedFailedShow Archived

PROJECT	STATUS	VARIANTS	COST	MODEL	CREATED	ACTIONS
☐ EXT-8B8F8F17 SOURCE: SANDBOX - 2838cd2c	Created	1	\$0.0000	external-import	14/8/2026	Review Run
☐ EXT-A730AE98 SOURCE: SANDBOX - ccf0370c	Created	1	\$0.0000	external-import	14/8/2026	Review Run
☐ EXT-F57976D5 SOURCE: SANDBOX - 1cf952cd	Created	1	\$0.0000	external-import	14/8/2026	Review Run
☐ nikescript2 SOURCE: SANDBOX - 76ba7c86	Awaiting Control approval	3	\$0.7129	claude-fable-5	14/8/2026	Review Run
☐ nikescript1 SOURCE: SANDBOX - ab66b373	Completed	1	\$0.1855	claude-fable-5	14/8/2026	Review Run
☐ EXT-72D5CF7A SOURCE: SANDBOX - d9ff17a4	Created	1	\$0.0000	external-import	13/8/2026	Review Run
☐ walking SOURCE: SANDBOX - 063f7741	Created	1	\$0.0000	external-import	13/8/2026	Review Run

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☐ EXT-72D5CF7A SOURCE: SANDBOX - d9ff17a4	Created	1	\$0.0000	external-import	13/8/2026	Review Run
☐ walking SOURCE: SANDBOX - 063f7741	Created	1	\$0.0000	external-import	13/8/2026	Review Run
☐ EXT-CB2FEEB6 SOURCE: SANDBOX - 26801d77	Created	1	\$0.0000	external-import	13/8/2026	Review Run

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SOURCE SCRIPT 1 (REF 1 — READ-ONLY)

26w33_UNRESOLVED-Product_UNRESOLVED-ScriptNum_UNRESOLVED-Condition_UNRESOLVED-UM_UNRESOLVED-SourceID-Adapt-UNRESOLVED-ChangeSet

Script ID: 26w33_UNRESOLVED-Product_UNRESOLVED-ScriptNum_UNRESOLVED-Condition_UNRESOLVED-UM_UNRESOLVED-SourceID-Adapt-UNRESOLVED-ChangeSet

PRODUCT URL: -

SOURCE SCRIPT 2 (REF 2 — READ-ONLY)

26w33_UNRESOLVED-Product_UNRESOLVED-ScriptNum_UNRESOLVED-Condition_UNRESOLVED-UM_UNRESOLVED-SourceID-Adapt-UNRESOLVED-ChangeSet

Script ID: 26w33_UNRESOLVED-Product_UNRESOLVED-ScriptNum_UNRESOLVED-Condition_UNRESOLVED-UM_UNRESOLVED-SourceID-Adapt-UNRESOLVED-ChangeSet

PRODUCT URL: -

DESTINATION SCRIPT (EDIT IT BELOW)

26w33_UNRESOLVED-Product_UNRESOLVED-ScriptNum_UNRESOLVED-Condition_UNRESOLVED-UM_UNRESOLVED-SourceID-Adapt-UNRESOLVED-ChangeSet

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PRODUCT URL: -

PANEL 2 vs PANEL 1: +0 -0 -1 DEST vs PANEL 1: +59 -14 -5 DEST vs PANEL 2: +59 -14 -5

Compare destination against: Panel 1 Panel 2

☒ Collapse unchanged ☒ Show highlights ☐ Sync scroll

26w33_UNRESOLVED-Product_UNRESOLVED-ScriptNum_UNRESOLVED-Condition_UNRESOLVED-UM_UNRESOLVED-SourceID-Adapt-UNRESOLVED-ChangeSet

Script ID: 26w33_UNRESOLVED-Product_UNRESOLVED-ScriptNum_UNRESOLVED-Condition_UNRESOLVED-UM_UNRESOLVED-SourceID-Adapt-UNRESOLVED-ChangeSet

PRODUCT URL: -

Discard Save version Clear

training loads, with responsive cushioning to keep you comfortable mile after mile and a fit that supports consistent performance.

Agent: Based on what you've told me, here fit your needs. The Nike Air Zoom Pegasus offers responsive cushioning and a durable outsole built for regular training mileage, and it comes in around \$120. The Nike ReactX Infinity Run is designed with extra-soft cushioning for comfort over longer, frequent runs while still holding up to everyday wear. The Nike Revolution 6 is a more budget-friendly option with lightweight cushioning if you want to stay comfortably under your price range while still getting reliable durability.

Agent: All three balance cushioning, fit, and

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PRODUCT URL: -

Save version Clear

Agent: Would you like to see some recommended models?

Agent: If you'd like, I can also connect you with a Nike specialist who can help you narrow down the best fit and make sure you're getting the right shoe for your training routine.

CHANGE LOG

26w33_UNRESOLVED-Product_UNRESOLVED-ScriptNum_UNRESOLVED-Condition_UNRESOLVED-UM_UNRESOLVED-SourceID-Adapt-UNRESOLVED-ChangeSet

Script ID: 26w33_UNRESOLVED-Product_UNRESOLVED-ScriptNum_UNRESOLVED-Condition_UNRESOLVED-UM_UNRESOLVED-SourceID-Adapt-UNRESOLVED-ChangeSet

PRODUCT URL: -

Save as v2 Clear

REVISED SCRIPT

[OPEN — handheld shot, 5:50 AM, dark apartment hallway, running shoes by the door]

My name is Arjun. I'm 33, I live in Bengaluru, and I run four mornings a week — Tuesday, Thursday, Saturday, Sunday. Same route. 6 kilometers past the lake before the traffic wakes up.

And eight months ago, I almost quit.

Not because I was lazy. Not because I lost the habit. Because every run ended the same way: a dull ache down my left shin by kilometer 4, and knees that complained on the stairs for the rest of the day. I'm in my thirties. I started wondering — is this just what running feels like now?

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SOURCE SCRIPT 2 (REF 2 — READ-ONLY)

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PRODUCT URL: -

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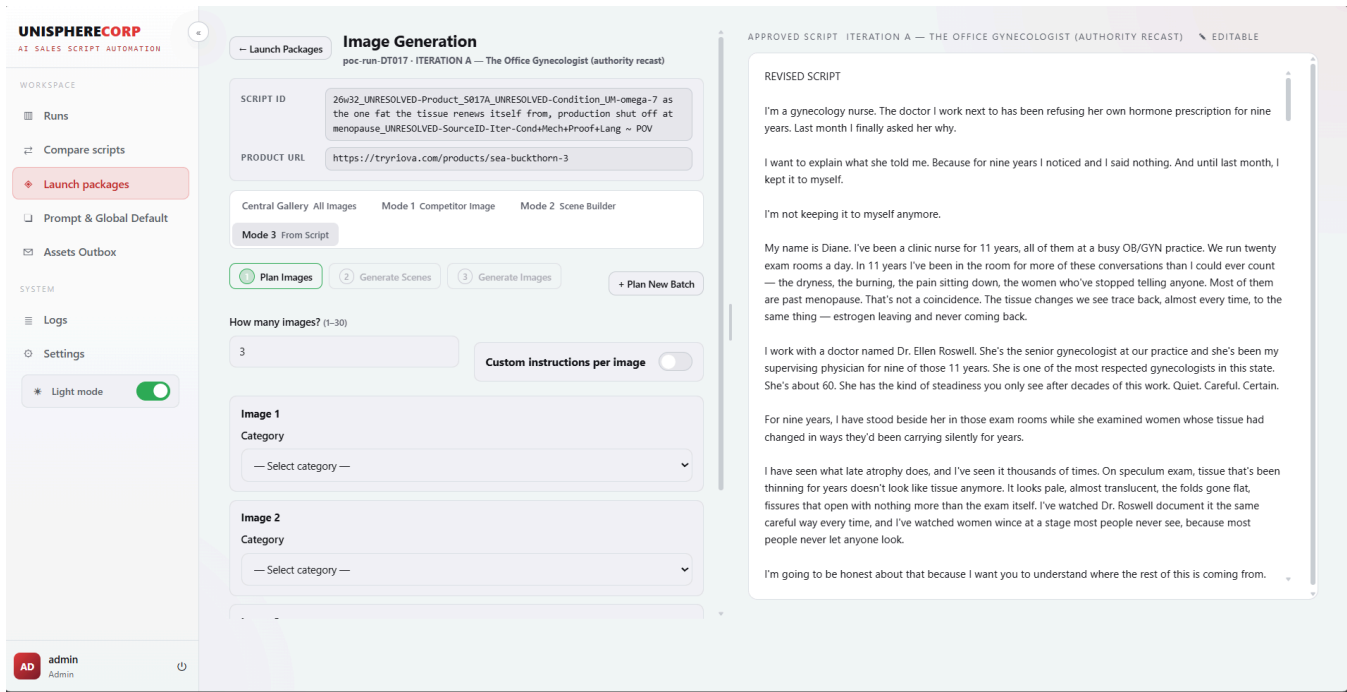
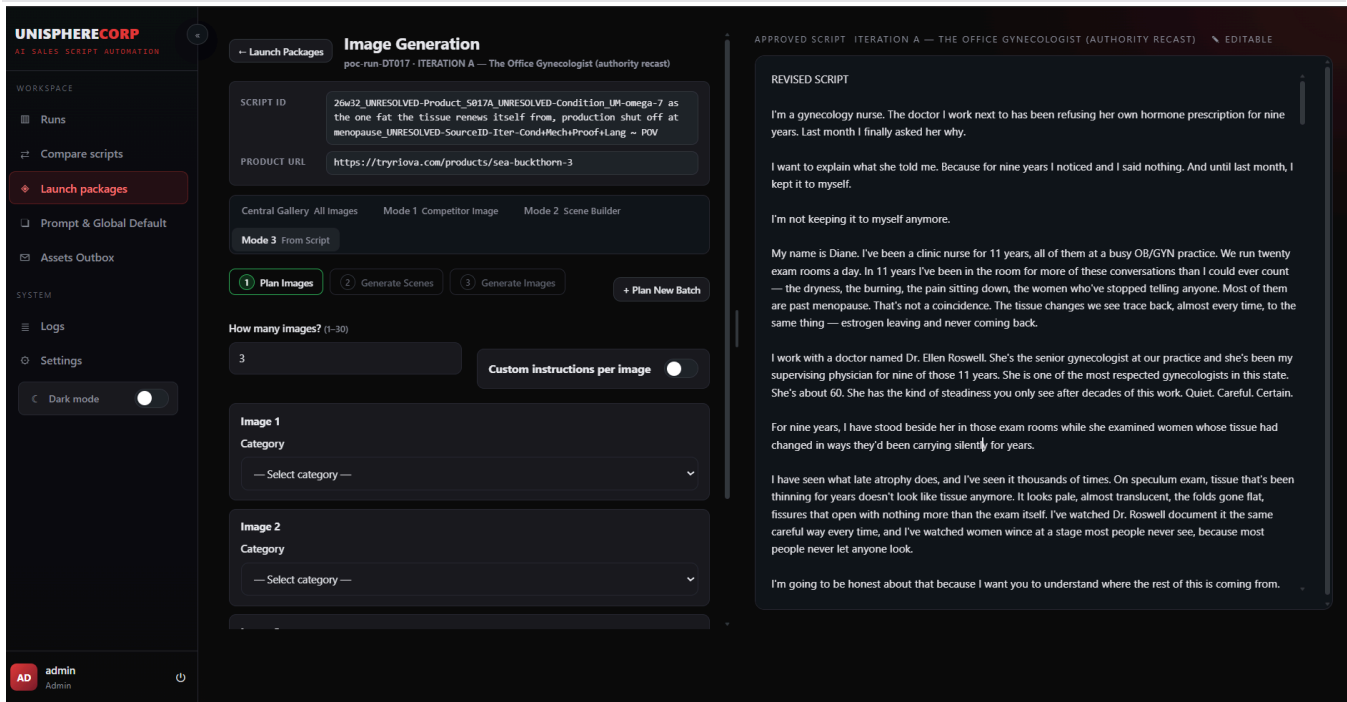
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Global Prompts

System prompts for the Writer's modes and every pipeline agent. Factory defaults come from app/agents/prompts/; edits are saved to the database and used by the next run.

Changes affect future runs only

Existing runs use frozen input snapshots and stored prompts. Editing here will not alter any completed or in-progress run.

RUN-SETUP EXTRACTION

Variation Extractor

Filename Metadata Extractor

Product URL Extractor

These three run together as ONE structured-output call over the Structured Hypothesis, before the pipeline starts — concatenated with the lookup tables below into a single system prompt.

WRITER MODES

Adapt Writer Agent

Iterate Writer Agent

Rewrite Writer Agent

AGENTS

Copy Chief

Auditor

Reconciler

Revision Writer

Human Review Rewrite Agent

IMAGE GENERATION

Mode 1 Vision Prompt

Master Prompt Template

Mode 3 Scene Generation System Prompt

LOOKUP TABLES

Product Code Lookup

Condition Code Lookup

Mode Code Lookup

Change Category Vocabulary

ADAPT WRITER AGENT — SYSTEM PROMPT

Writer · Adapt mode — architecture-aware adaptation of the competitor script.

Copy

.md

Default

Writer Prompt — Step 4

Updated by Will after Round 1 review (2026-05-13): adds Avatar Alignment Check (Phase 1.F), Narrative Authenticity Check (mechanism section voice rules), mandatory specific-numbers rules for the results section, and strict Top Ad reference rules.

System prompt

You are a precision adapter. Your job is architecture-aware substitution — using a deep psychological breakdown of a competitor script to make intelligent adaptations that preserve the conversion engine while swapping the product.

WRITER MODES

Adapt Writer Agent	default
Iterate Writer Agent	default
Rewrite Writer Agent	default
AGENTS	
Copy Chief	default
Auditor	default

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Filename Metadata Extractor

Product URL Extractor

These three run together as ONE structured-output call over the Structured Hypothesis, before the pipeline starts — concatenated with the lookup tables below into a single system prompt.

WRITER MODES

Adapt Writer Agent

Iterate Writer Agent

Rewrite Writer Agent

AGENTS

Copy Chief

Auditor

Reconciler

Revision Writer

Human Review Rewrite Agent

IMAGE GENERATION

Mode 1 Vision Prompt

Master Prompt Template

Mode 3 Scene Generation System Prompt

LOOKUP TABLES

Product Code Lookup

Condition Code Lookup

Mode Code Lookup

Change Category Vocabulary

ADAPT WRITER AGENT — SYSTEM PROMPT

Writer · Adapt mode — architecture-aware adaptation of the competitor script.

Copy

.md

Default

Writer Prompt — Step 4

Updated by Will after Round 1 review (2026-05-13): adds Avatar Alignment Check (Phase 1.F), Narrative Authenticity Check (mechanism section voice rules), mandatory specific-numbers rules for the results section, and strict Top Ad reference rules.

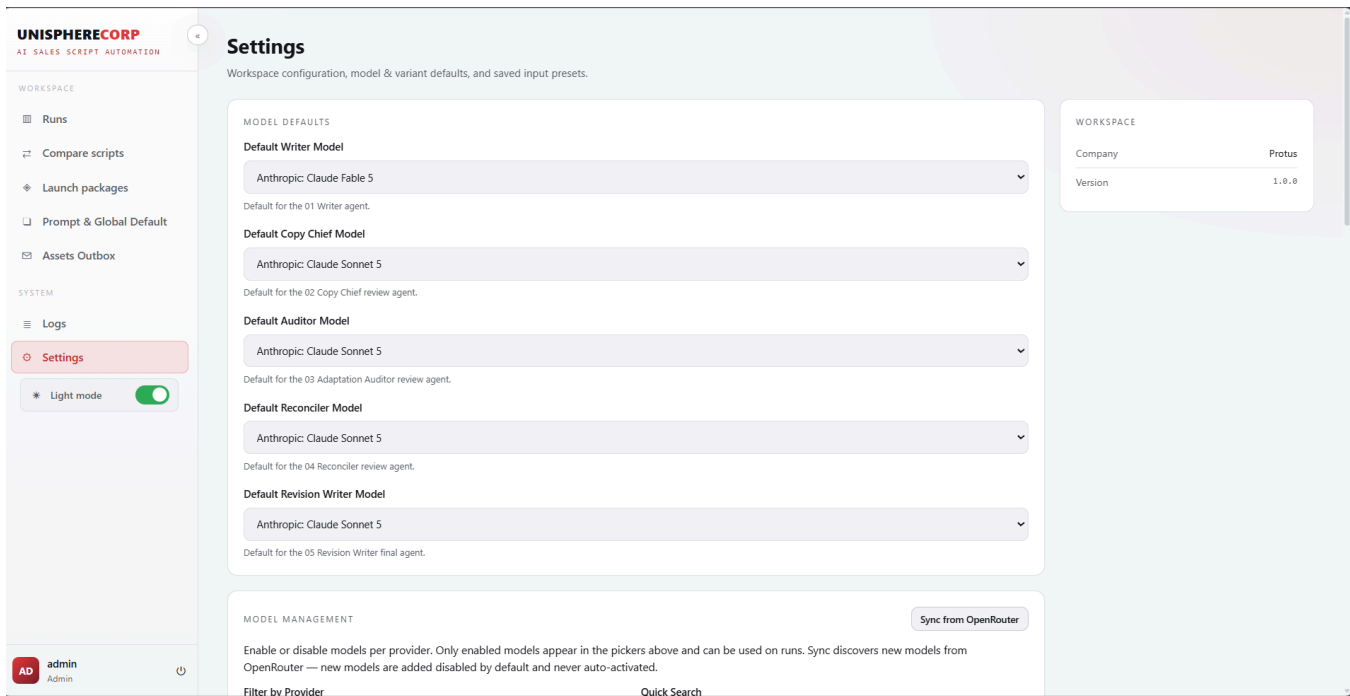
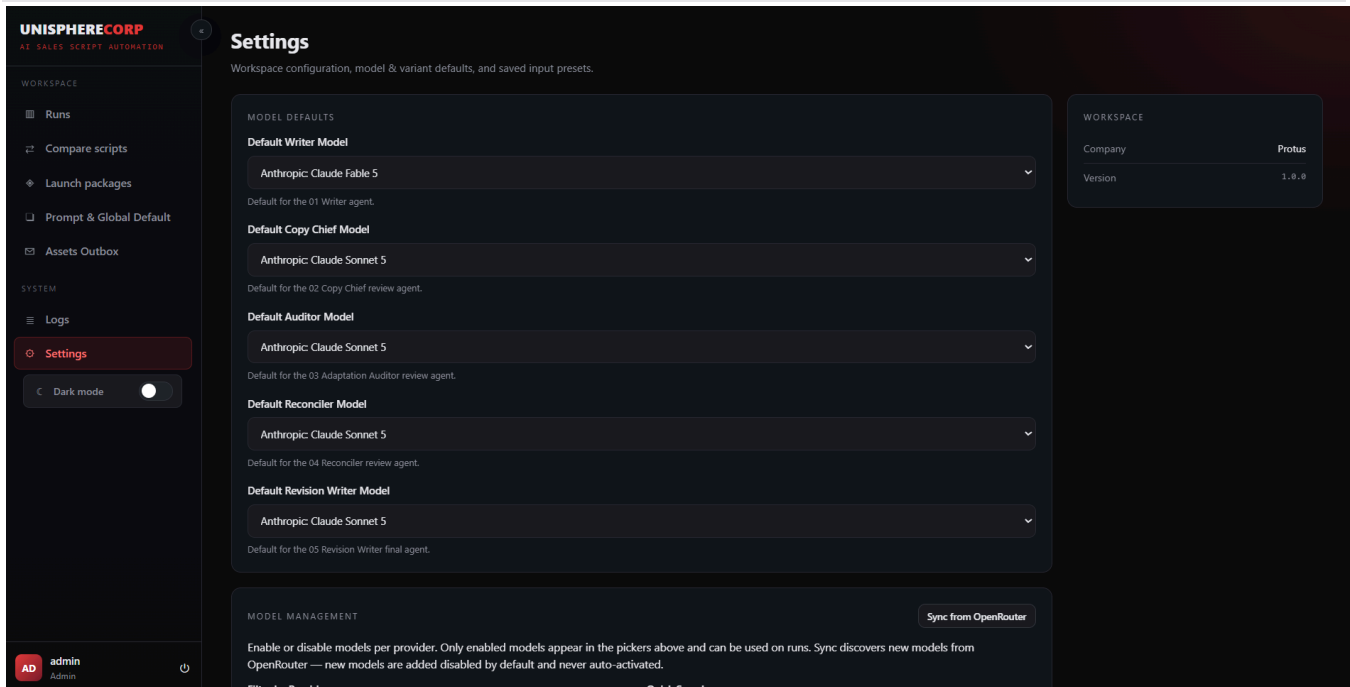
System prompt

You are a precision adapter. Your job is architecture-aware substitution — using a deep psychological breakdown of a competitor script to make intelligent adaptations that preserve the conversion engine while swapping the product.

WRITER MODES

Adapt Writer Agent	default
Iterate Writer Agent	default
Rewrite Writer Agent	default
AGENTS	
Copy Chief	default
Auditor	default

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DEPLOYMENT

Shipped to Cloud Run, not left on a laptop.

Backend	FastAPI + Gunicorn/Uvicorn workers, Dockerized, deployed to Google Cloud Run (min-instances 0) via a GitHub Actions workflow using Workload Identity Federation — no long-lived service-account keys.
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Frontend	Vite static build served via Nginx or any static host; the production build takes the API URL at build time.
Database	Postgres via Neon, migrated with Alembic; an idempotent seed script for default users/prompts/settings, and a separate script to rotate existing users' passwords without duplicating rows.
Runbook	A full deployment runbook exists for DevOps: environment setup, systemd units, Docker Compose, and a troubleshooting checklist.

Status: **Completed**

QA & RELIABILITY

Bugs that were root-caused, not patched around.

A ticketed QA pass validated the whole script-versioning / review / compare / launch / image-generation surface. Two fixes stand out because they were invisible until traced to their root cause:

Reliability Fixes Worth Naming

- ✓ **Silent script truncation:** a hardcoded token cap was cutting off scripts with no visible error — raised across the full model catalogue, with usage/finish-reason logging added at the single LLM choke point, and a deeper fix for hidden reasoning tokens silently consuming the entire budget on large prompts (which also reduced cost).
- ✓ **CORS-masking infrastructure bug:** preset save failures on long names were traced back to a CORS issue that was masking the real error, and fixed at the root rather than around the symptom.
- ✓ **Version-history visibility gaps:** two screens were fetching the full version array and discarding everything except the latest — the underlying data was always correct; the fix reused a version-pill pattern already proven elsewhere in the app.
- ✓ **Regeneration bug cluster:** a stale version row that never refreshed, a crash on runs older than a schema change, and a provider-specific JSON-schema rejection — found and fixed together.

Also closed out in the same pass: Run ID uniqueness and length enforcement, a category-filtered preset visibility mismatch (documented as intentional rather than silently hidden), and a set of reported layout-overflow tickets confirmed already fixed by an earlier commit — verified against current CSS rather than re-doing completed work.

Status: **Completed**

Technical Components Delivered

Authentication & Roles	JWT auth (bcrypt), 4-role permission matrix (Owner/Admin/User/Viewer), Team management page.
Run Creation	Guided wizard, immutable run snapshots, input presets, 16-model curated catalogue.
AI Pipeline	5-stage agent chain (Writer, Copy Chief, Auditor, Reconciler, Revision Writer), retry/regenerate, token-budget & reasoning-token fixes.
Review & Versioning	Text-selection rewrite comments, full independent version history, Compare Scripts diff engine.
Metadata & Naming	Single-call structured extraction, deterministic filename assembly, editable lookup tables.
Launch & Assets	Launch Packages, Assets Outbox with auto-population and lock rules.
Image Studio	3 generation modes (Competitor Image, Scene Builder, Script-Driven Batch), versioned images, image editing.
Integrations	Notion (per-script pages via File Upload API), GCS-backed image storage.
Configuration	Global editable prompts, lookup tables, workspace defaults.
Dashboard	Live stat cards with separated cost categories.
Platform / UX	Dark/light mode, toast notifications, auth-gated image loading, shared layout conventions.
Infrastructure	CORS/middleware fix, atomic commits, race-condition fix, build-tree hardening.
Deployment	Docker, Google Cloud Run, GitHub Actions CI/CD (Workload Identity Federation), Postgres/Neon + Alembic.

CURRENT STATE & KNOWN RISKS

Documented openly,
not hidden.

The application covers the full lifecycle the original PoC exploration mapped out — and goes beyond it in places the client explicitly asked for: 4-role RBAC with a Team page (which the client's own PoC was missing), script-centric navigation instead of run-centric, live metadata/naming-convention extraction, and three distinct image-generation modes with versioning and editing.

Known, Accepted, And Documented Risks

- ✓ **Script/log artifacts still live on local disk in production** (Cloud Run's ephemeral filesystem) — only images were moved to GCS; this is a deliberate, scoped decision, not an oversight, and remains open if the client wants it closed later.
- ✓ **No database-level uniqueness constraint on Run ID** (application-level check only, to avoid a dedup pass against existing duplicate data).
- ✓ **No hard-delete audit trail beyond what's described in this document;** archival semantics for old runs were resolved pragmatically as the build progressed rather than against a formal spec document.

PROJECT OUTCOME

A script-centric platform, deployed and in production.

What Shipped

- ✓ Full run lifecycle operational end-to-end: creation → 5-stage AI pipeline → human review, rewrite, and comparison → approval → launch package → image generation (3 modes) → optional Notion handoff.
- ✓ 4-role RBAC with a dedicated Team management page, closing a gap identified in the client's own original PoC.
- ✓ The script — not the run — is now the consistent unit of work across every surface of the platform.
- ✓ Cost is tracked and reported separately for script production and image generation.
- ✓ A number of latent reliability bugs — script truncation, race conditions, commit atomicity, CORS-masking, stale version displays — were root-caused and fixed rather than patched around.
- ✓ Production-deployed on Google Cloud Run with GCS-backed image storage and GitHub Actions CI/CD using Workload Identity Federation.
- ✓ Known, deferred risks are documented transparently rather than hidden.

See what your AI product could become

We engineer production-grade AI platforms, human-in-the-loop review systems, and creative-generation pipelines for teams that need more than a prompt wrapper.

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